

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1260

Adrian 138 kV
August 25, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1260. The Interconnection Customer proposes a 333.3 MW natural gas combined cycle facility interconnecting at the Adrian 138 kV substation in the New Jersey area. The request was received on July 28, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 318 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$360,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: ADRIAN - WHITE TANKS SRP 138kV
Contingency: ADRIAN - HALESITE 345kV
Loading: 135.6% (Rating: 287 MVA)
DFAX: 36.3%
Pre-project loading: 93.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ADRIAN - WHITE TANKS SRP 138kV	Rebuild line with higher-capacity conductor	\$664,000	54% (\$360,000)
	Total allocated to project		\$360,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.